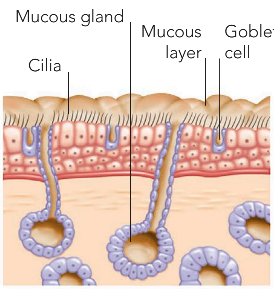
CHRONIC OBSTRUCTIVE PULMONARY DISEASE

Chronic obstructive pulmonary disease (COPD) consists primarily of chronic bronchitis and emphysema, two conditions that usually occur together. In this disorder there is progressive damage to lung tissue, causing restricted airflow in and out of the lungs and shortness of breath.

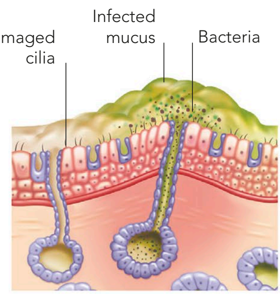
CHRONIC BRONCHITIS

In chronic bronchitis, the main airways leading to the lungs, the bronchi, become inflamed, congested, and narrowed due to irritation caused by tobacco smoke, frequent infections, or prolonged exposure to pollutants. The inflamed airways begin to produce too much mucus (sputum), resulting in a typical cough that at first is troublesome mostly in damp, cold months but then persists throughout the year. Symptoms such as hoarseness, wheezing, and breathlessness also develop. Eventually a person becomes short of breath even at rest. If a secondary respiratory infection develops, the sputum may change appearance from clear or white to yellow or green.

NORMAL AIRWAY LINING
Glands produce mucus that traps inhaled dust and germs. Tiny surface hairs (cilia) propel the mucus up into the throat, where it is coughed up or swallowed.



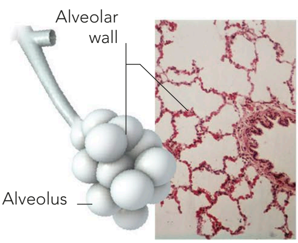
AIRWAY IN CHRONIC BRONCHITIS
Inhaled irritants cause glands to produce more mucus. Damaged cilia cannot propel mucus along, so it becomes a bacterial breeding ground.



EMPHYSEMA

In emphysema, the air sacs (alveoli) become overstretched. They rupture and merge, which reduces their oxygen-absorbing surfaces and makes gas exchange less efficient. Air also becomes trapped inside them, the lungs over-inflate, and the volume of air moving in and out of the lungs is reduced. Most people affected by emphysema are long-term heavy smokers, although a rare inherited condition called alpha1-antitrypsin deficiency can also cause the disorder. The lung damage is usually irreversible, but giving up smoking may slow the progression of the disease.

HEALTHY TISSUE
The alveoli are grouped, like grapes, and each tiny sac is partly separate from the others. The walls are thin and elastic so they can stretch.



DAMAGED TISSUE
Smoke or other pollutants stimulate chemicals that cause the alveolar walls to break down, reducing the area for gas exchange.